A05_DataVisualization

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Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
#install.packages("cowplot")
library(ggplot2)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```r
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
lake <- read.csv(here("Data", "Processed_KEY",
                  "NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
stringsAsFactors = T)
litter <- read.csv(here("Data", "Processed_KEY",
"NEON_NIWO_Litter_mass_trap_Processed.csv"), stringsAsFactors = T)
library(lubridate)
#2
glimpse(lake)
```

```
## Rows: 23,008
## Columns: 15
## $ lakename        <fct> Paul Lake, Paul Lake, Paul Lake, Paul Lake, Paul Lake,~
## $ year4           <int> 1984, 1984, 1984, 1984, 1984, 1984, 1984, 1984, 1984, ~
## $ daynum          <int> 148, 148, 148, 148, 148, 148, 148, 148, 148, 148, 148,~
## $ month           <int> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, ~
## $ sampledate      <fct> 1984-05-27, 1984-05-27, 1984-05-27, 1984-05-27, 1984-0~
## $ depth           <dbl> 0.00, 0.25, 0.50, 0.75, 1.00, 1.50, 2.00, 3.00, 4.00, ~
## $ temperature_C   <dbl> 14.5, NA, NA, NA, 14.5, NA, 14.2, 11.0, 7.0, 6.1, 5.5,~
## $ dissolvedOxygen <dbl> 9.5, NA, NA, NA, 8.8, NA, 8.6, 11.5, 11.9, 2.5, 1.6, 0~
## $ irradianceWater <dbl> 1750.0, 1550.0, 1150.0, 975.0, 870.0, 610.0, 420.0, 22~
## $ irradianceDeck  <dbl> 1620, 1620, 1620, 1620, 1620, 1620, 1620, 1620, 1620, ~
## $ tn_ug           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ tp_ug           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ nh34            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ no23            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ po4             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
```

```r
glimpse(litter)
```

```
## Rows: 1,692
## Columns: 13
```

```
## $ plotID           <fct> NIWO_062, NIWO_061, NIWO_062, NIWO_064, NIWO_058, NIW~
## $ trapID           <fct> NIWO_062_050, NIWO_061_169, NIWO_062_050, NIWO_064_10~
## $ collectDate      <fct> 2016-06-16, 2016-06-16, 2016-06-16, 2016-06-16, 2016-~
## $ functionalGroup  <fct> Seeds, Other, Woody material, Seeds, Needles, Leaves,~
## $ dryMass          <dbl> 0.000, 0.270, 0.120, 0.000, 1.110, 0.000, 0.000, 0.00~
## $ qaDryMass        <fct> N, N, N, N, Y, N, N, N, N, N, N, Y, N, N, N, N, N, Y,~
## $ subplotID        <int> 31, 41, 31, 32, 32, 32, 40, 40, 40, 40, 40, 31, 31, 3~
## $ decimalLatitude  <dbl> 40.05114, 40.04762, 40.05114, 40.04737, 40.04872, 40.~
## $ decimalLongitude <dbl> -105.5858, -105.5861, -105.5858, -105.5840, -105.5872~
## $ elevation        <dbl> 3477.0, 3413.4, 3477.0, 3373.2, 3446.4, 3446.4, 3509.~
## $ nlcdClass        <fct> shrubScrub, evergreenForest, shrubScrub, evergreenFor~
## $ plotType         <fct> tower, tower, tower, tower, tower, tower, tower, towe~
## $ geodeticDatum    <fct> WGS84, WGS84, WGS84, WGS84, WGS84, WGS84, WGS84, WGS8~
```

```
lake$sampledate <- ymd(lake$sampledate)
lake$month <- factor(lake$month,levels=1:12,labels=month.abb)
litter$collectDate <- ymd(litter$collectDate)
#date is expressed correctly
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
theme_Liam <- function() {theme_minimal(base_size = 12, base_family = "Helvetica") +
    theme(plot.title = element_text(face = "bold", size = 16, hjust
    = 0.5, color = "blue"), axis.title =
    element_text(size = 13, face = "bold", color = "red"))}
theme_set(theme_Liam())
#kept it simple with the colors and elements, might add more to default
#with time
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
head(lake)
```

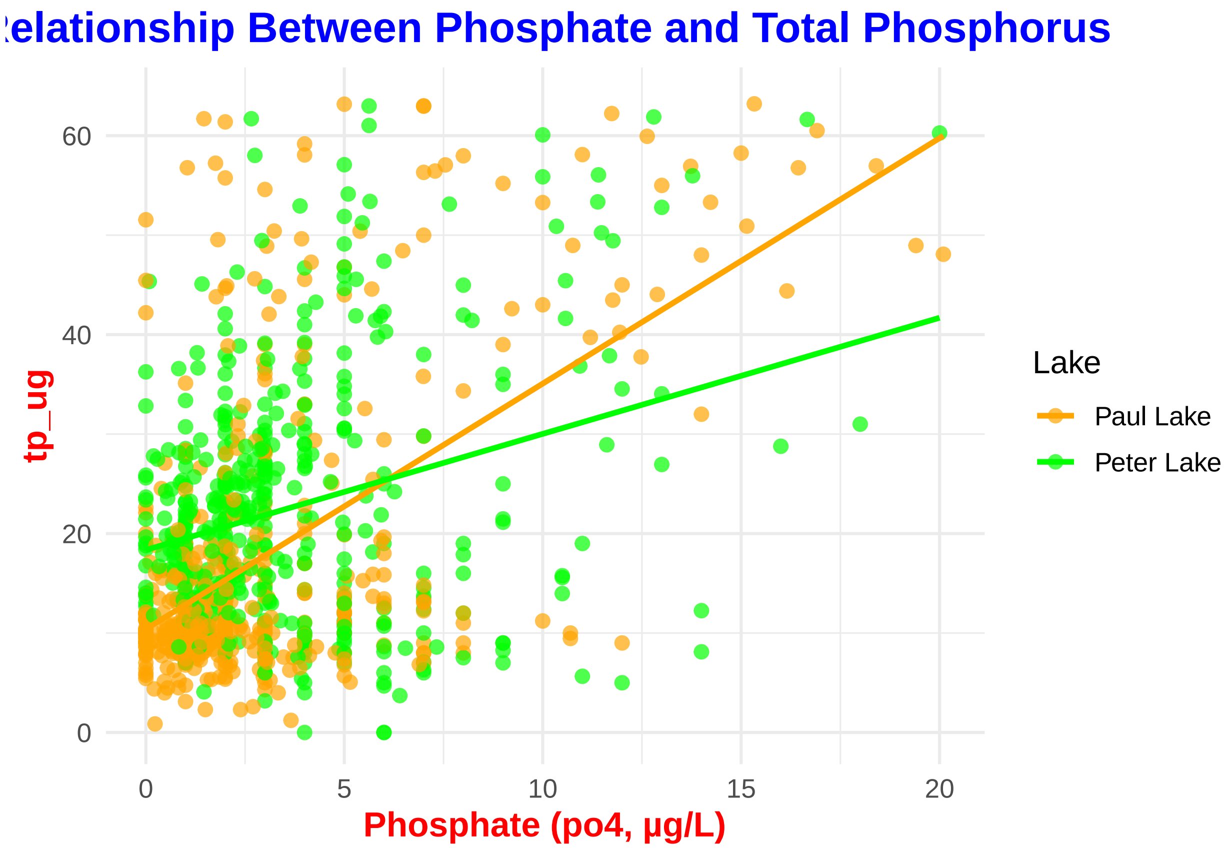
```
##    lakename year4 daynum month sampledate depth temperature_C dissolvedOxygen
## 1 Paul Lake  1984    148   May 1984-05-27  0.00          14.5             9.5
## 2 Paul Lake  1984    148   May 1984-05-27  0.25            NA              NA
## 3 Paul Lake  1984    148   May 1984-05-27  0.50            NA              NA
## 4 Paul Lake  1984    148   May 1984-05-27  0.75            NA              NA
## 5 Paul Lake  1984    148   May 1984-05-27  1.00          14.5             8.8
## 6 Paul Lake  1984    148   May 1984-05-27  1.50            NA              NA
##   irradianceWater irradianceDeck tn_ug tp_ug nh34 no23 po4
## 1            1750           1620    NA    NA   NA   NA  NA
## 2            1550           1620    NA    NA   NA   NA  NA
## 3            1150           1620    NA    NA   NA   NA  NA
## 4             975           1620    NA    NA   NA   NA  NA
## 5             870           1620    NA    NA   NA   NA  NA
## 6             610           1620    NA    NA   NA   NA  NA
```

```r
str(lake)
```

```
## 'data.frame':    23008 obs. of  15 variables:
##  $ lakename       : Factor w/ 2 levels "Paul Lake","Peter Lake": 1 1 1 1 1 1 1 1 1 1 ...
##  $ year4          : int  1984 1984 1984 1984 1984 1984 1984 1984 1984 1984 ...
##  $ daynum         : int  148 148 148 148 148 148 148 148 148 148 ...
##  $ month          : Factor w/ 12 levels "Jan","Feb","Mar",..: 5 5 5 5 5 5 5 5 5 5 ...
##  $ sampledate     : Date, format: "1984-05-27" "1984-05-27" ...
##  $ depth          : num  0 0.25 0.5 0.75 1 1.5 2 3 4 5 ...
##  $ temperature_C  : num  14.5 NA NA NA 14.5 NA 14.2 11 7 6.1 ...
##  $ dissolvedOxygen: num  9.5 NA NA NA 8.8 NA 8.6 11.5 11.9 2.5 ...
##  $ irradianceWater: num  1750 1550 1150 975 870 610 420 220 100 34 ...
##  $ irradianceDeck : num  1620 1620 1620 1620 1620 1620 1620 1620 1620 1620 ...
##  $ tn_ug          : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ tp_ug          : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ nh34           : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ no23           : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ po4            : num  NA NA NA NA NA NA NA NA NA NA ...
```

```r
ggplot(lake, aes(x = po4, y= tp_ug, color = lakename)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) +
  labs(
    title = "Relationship Between Phosphate and Total Phosphorus",
    x = "Phosphate (po4, µg/L)", color = "Lake") +
  xlim(0, quantile(lake$po4, 0.95, na.rm = TRUE)) +
  ylim(0, quantile(lake$tp_ug, 0.95, na.rm = TRUE)) +
  scale_color_manual(values = c("Peter Lake" = "green", "Paul Lake" =
  "orange")) + theme_Liam()
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 22012 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 22012 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



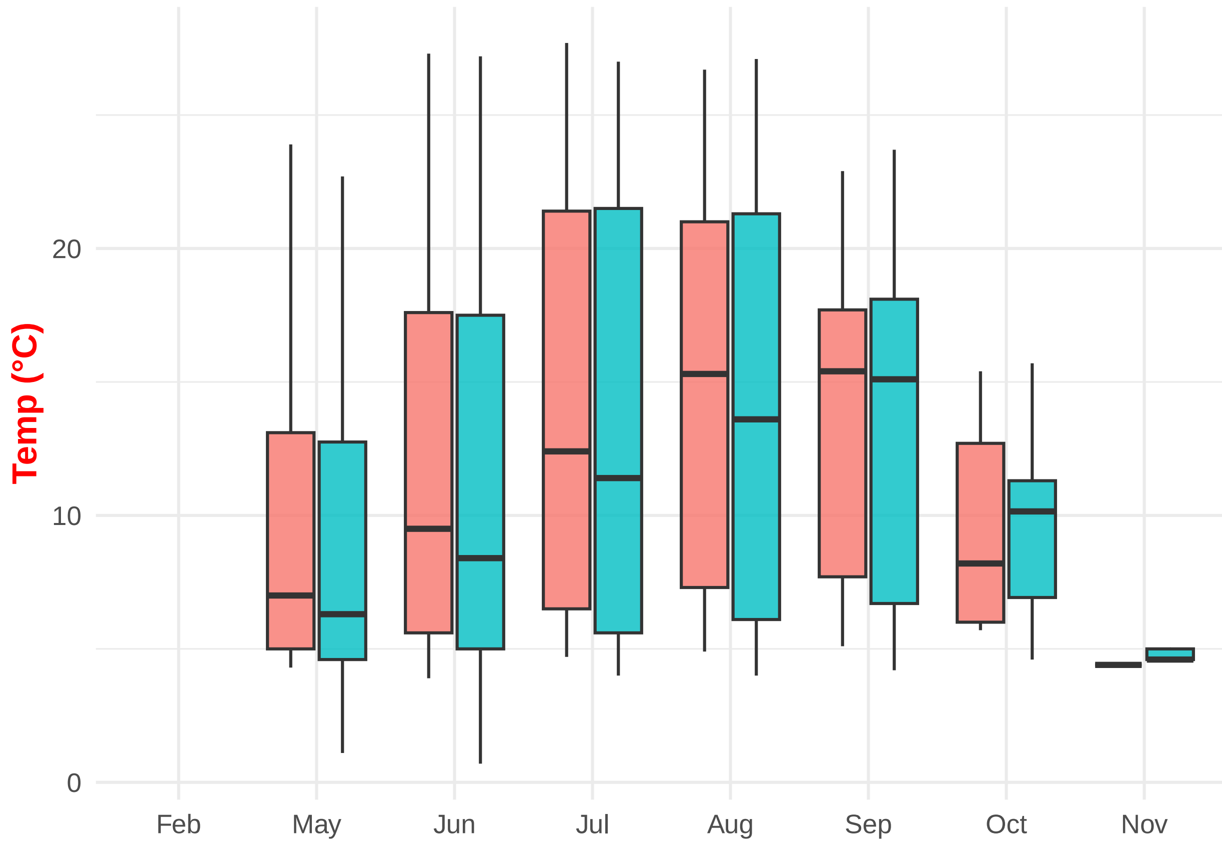

```
#above where color =, I had just lake and it caused the error
#coding still takes some getting used to
```

5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

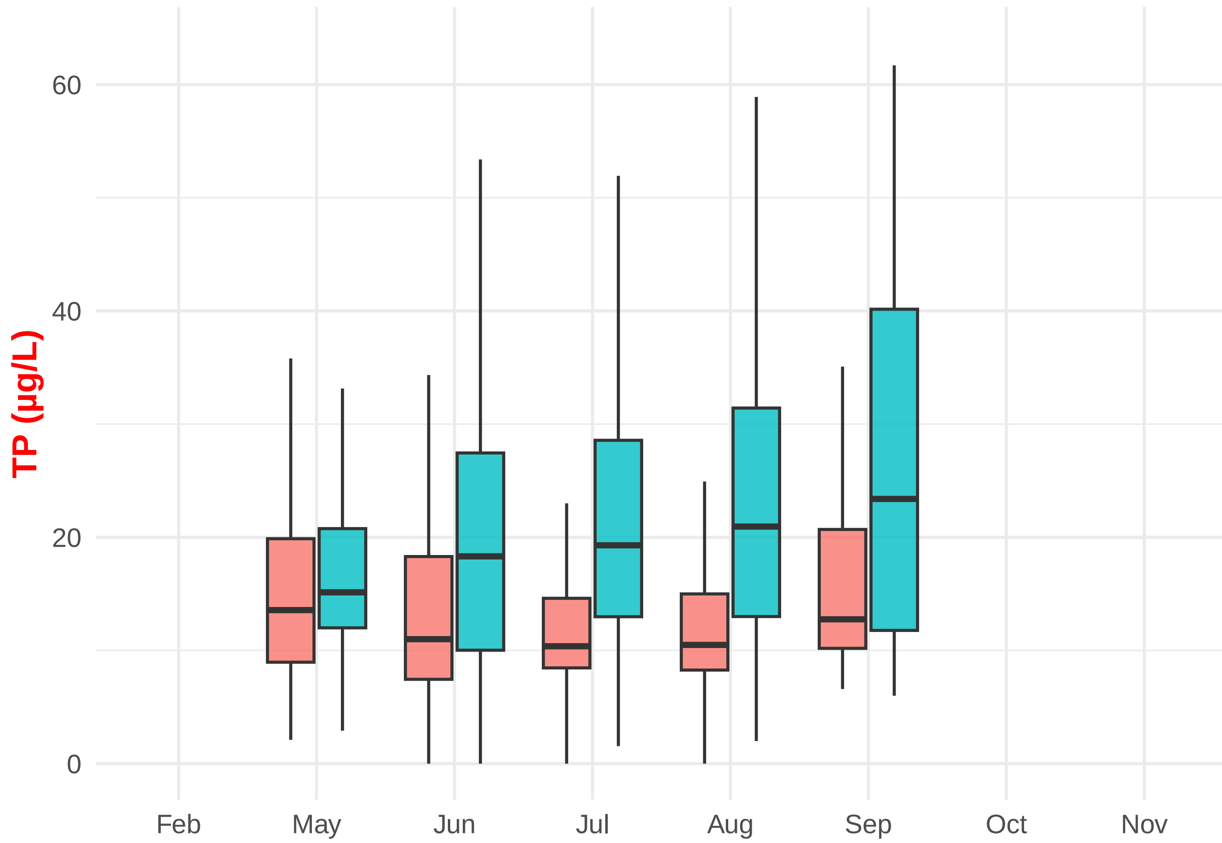
```
#5
p_temp <- ggplot(lake, aes(x = month, y = temperature_C, fill = lakename)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8) +
  labs(
    #title = "Temperature by Month",
    x = NULL,
    y = "Temp (°C)", fill = "lake"
  ) +
  theme_Liam() +
  theme(legend.position = "none")
print(p_temp)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```



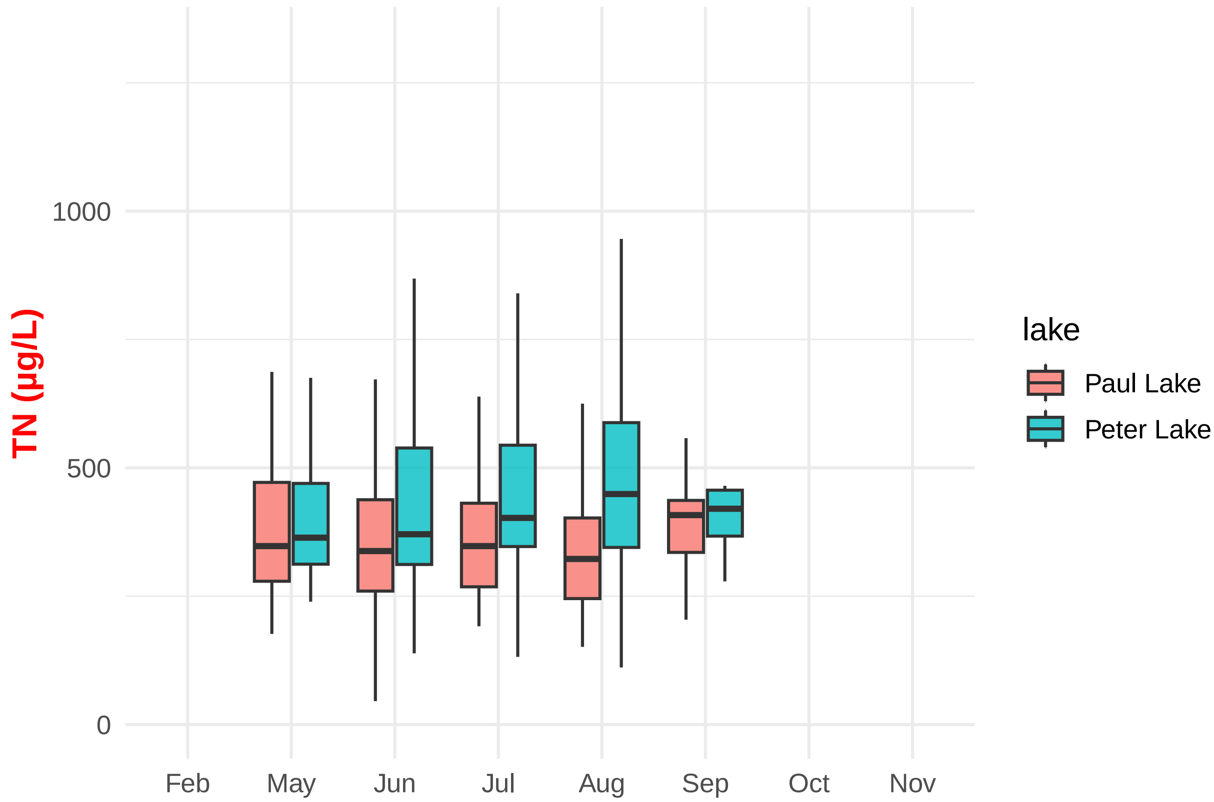

```
# first graph

p_tp <- ggplot(lake, aes(x = month, y = tp_ug, fill = lakename)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8) +
  labs(
    #title = "Total Phosphorus by Month",
    x = NULL,
    y = "TP (µg/L)", fill = "lake"
  ) + ylim(0, quantile(lake$tp_ug, 0.95, na.rm = TRUE)) +
  theme_Liam() +
  theme(legend.position = "none")
print(p_tp)
```

```
## Warning: Removed 20880 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
# graph 2
p_tn <- ggplot(lake, aes(x = month, y = tn_ug, fill = lakename)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8) +
  labs(
    #title = "Total Nitrogen by Month",
    x = "",
    y = "TN (µg/L)", fill = "lake"
  ) + ylim(0, quantile(lake$tn_ug, 0.95, na.rm = TRUE)) +
  theme_Liam()
print(p_tn)
```

```
## Warning: Removed 21655 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```



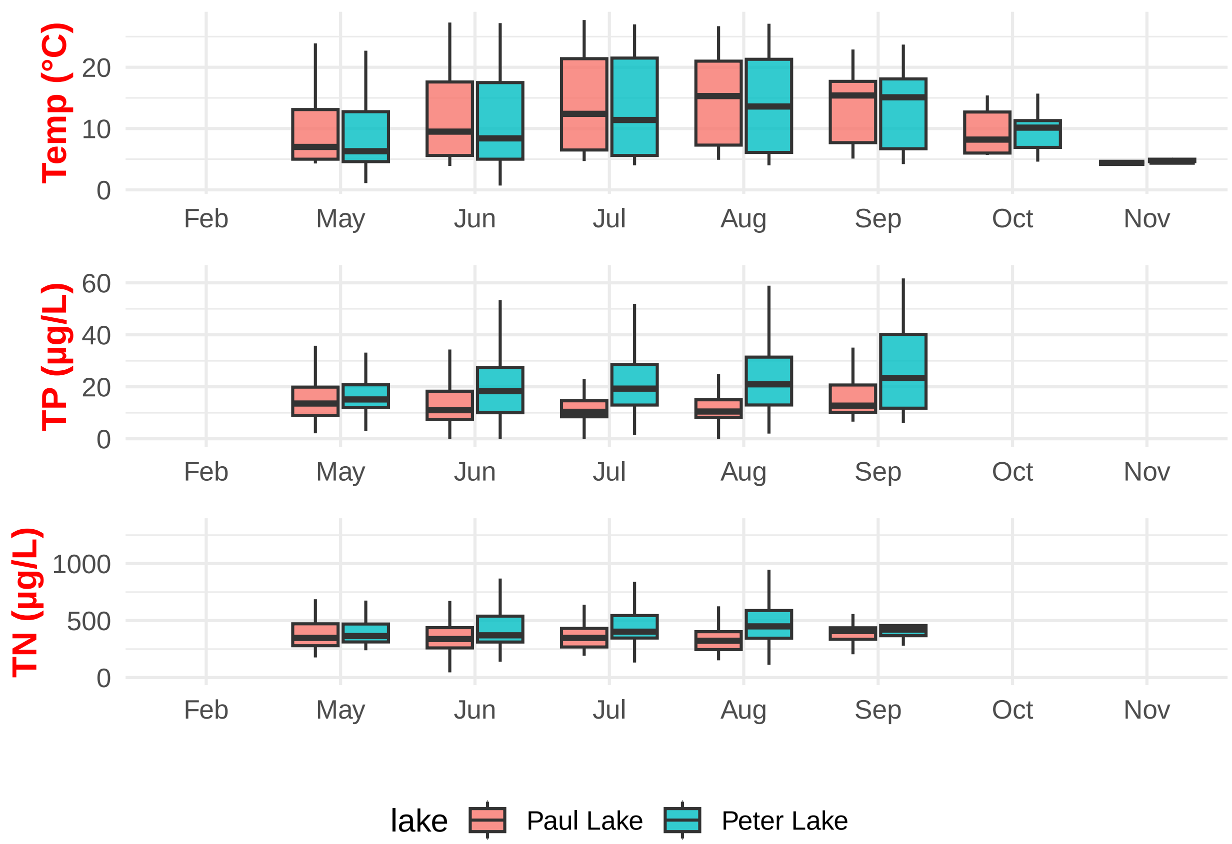

```
#graph 3, suprisingly this wasn't as hard as step 4
legend <- get_legend(p_tn + theme(legend.position = "bottom"))
```

```
## Warning: Removed 21655 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
combined_plots <- plot_grid(
  p_temp, p_tp, p_tn + theme(legend.position = "none"),
  ncol = 1,
  align = "v",
  axis = "lr",
  rel_heights = c(1, 1, 1.1)
)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 20880 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 21655 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```r
final_plot <- plot_grid(
  combined_plots, legend,
  ncol = 1,
  rel_heights = c(1, 0.1)
)
print(final_plot)
```




```r
#I think I did it right, it looks odd in the plot viewer??? There's
#something really strange where it only shows me months June-August
#in all of my plots, not sure why???
```

Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: I'm going to split my answer between the three variables. First, with temperature, its colder in the winter months and warmer in the non winter months. Between the two lakes, there is little to no difference in temperature, although Paul has a tiny bit higher of a median temperature. With total phosphorus, it peaks in the spring, the summer has some high levels although it varies depending on the lake and rises a little in September. Peter Lake also has has higher TP levels overall. Finally, with Total Nitrogen, these levels are also high in the spring but, drop after the summer ends. It appears they rise again in the fall. Once again, Peter lake has slightly higher concentrations in some months and a greater disparity in other months.

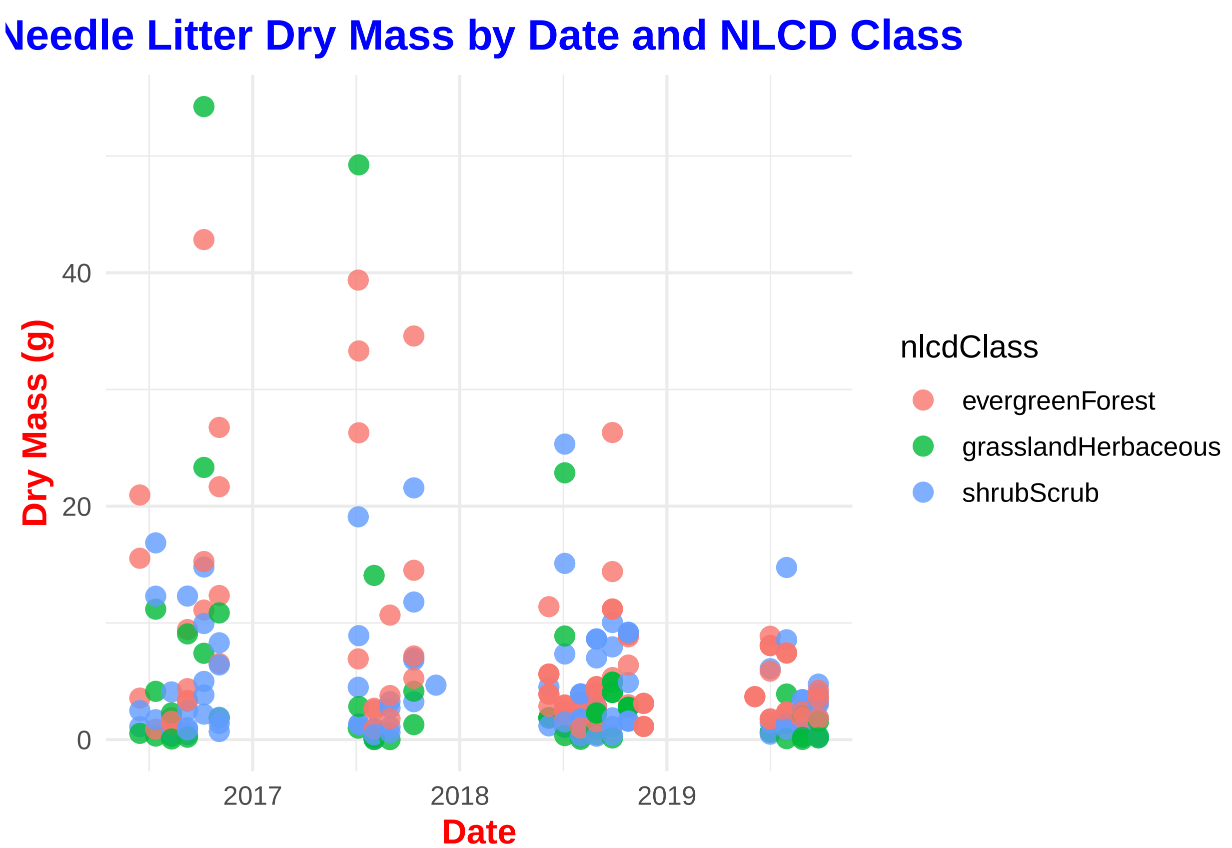
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)

7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
needles <- litter %>%
  filter(functionalGroup == "Needles")
unique(needles$functionalGroup)
```

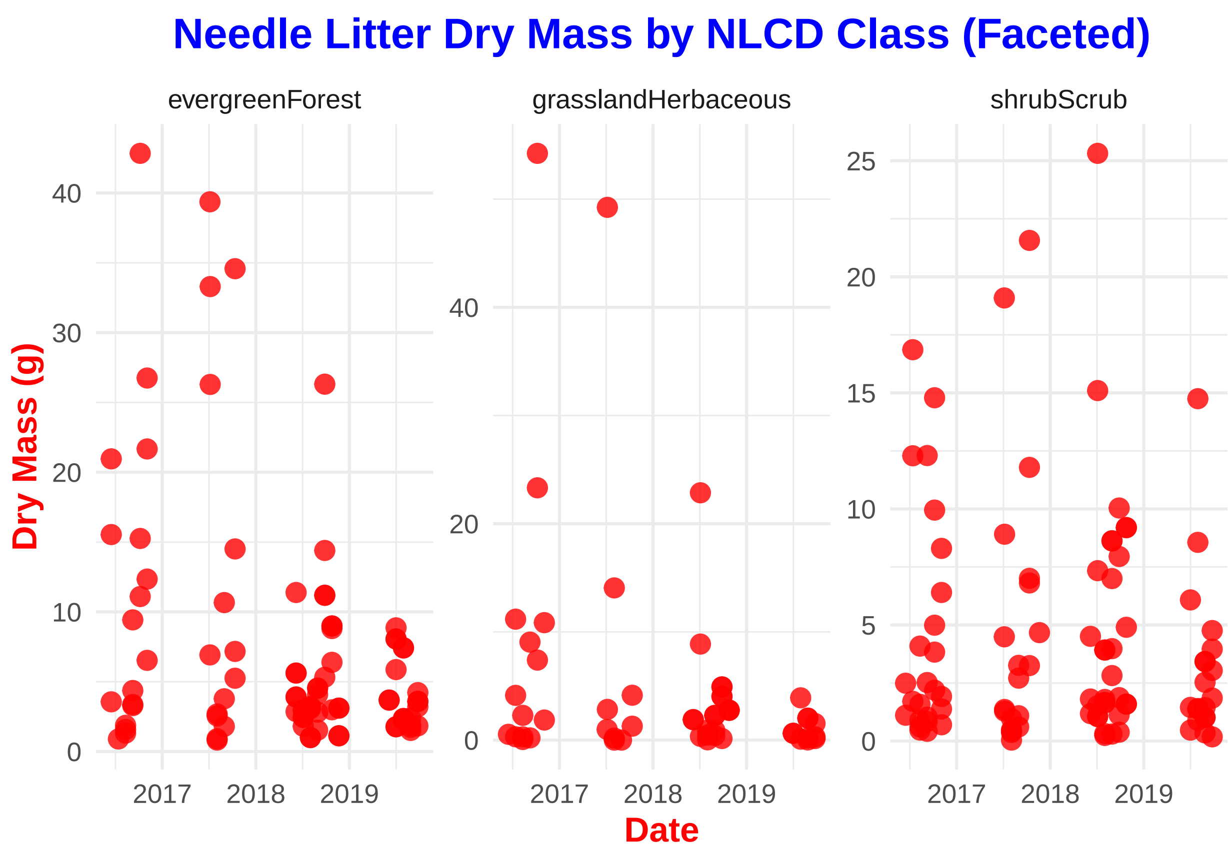
```
## [1] Needles
## 8 Levels: Flowers Leaves Mixed Needles Other Seeds ... Woody material
```

```
p_needles_color <- ggplot(needles, aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point(size = 3, alpha = 0.8) +
  #geom_line(aes(group = nlcdClass), linewidth = 1) +
  labs(title = "Needle Litter Dry Mass by Date and NLCD Class",
       x = "Date", y = "Dry Mass (g)") +
  theme_Liam()
print(p_needles_color)
```




```
#graph 1, I keep messing up the names of columns accidentally

#7
p_needles_facet <- ggplot(needles, aes(x = collectDate, y = dryMass)) +
  geom_point(color = "red", size = 3, alpha = 0.8) +
```

```
  #geom_line(linewidth = 1, color = "darkgreen") +
  facet_wrap(~nlcdClass, scales = "free_y") +
  labs(title = "Needle Litter Dry Mass by NLCD Class (Faceted)",
       x = "Date", y = "Dry Mass (g)") +
  theme_Liam()
print(p_needles_facet)
```




```
#second graph, I made the points red to be more distinguishable
```

Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: 7 is more effective as its easier to read and to discern the data between the different years. With plot 6, the data sits on top of each other and it is quite jumbled. With plot 7, while some of the data is still overlapping, I can clearly see the distinct metrics of dry mass for each ncld class.